

Related Literature: A Capital Shock and Chilean Manufacturing After the 2010 Maule Earthquake

Emily Liu

June 5, 2026

This review situates the project within four literatures it draws on and one it borrows its inference tools from. The organizing question is narrow: when a large, plausibly exogenous shock destroys and then rebuilds a region’s physical capital, which margin of the firm actually moves, and who captures the returns to reconstruction? Each strand below answers part of that question; none answers all of it at the establishment level, which is where this project sits.

1 Disasters and the trajectory of output

The first and largest strand measures what disasters do to aggregate or regional economic activity. The modern consensus is that large earthquakes leave persistent negative output footprints rather than the “creative destruction” rebound that an accelerated–replacement story might predict. [Aksoy et al. \[2024\]](#), assembling a global panel of earthquakes, find that output falls and stays below trend for years, with the damage concentrated in lower-income economies and those with weaker institutions; the capital-upgrading channel that could in principle raise productivity is empirically dominated by the destruction of the existing stock. For Chile specifically, [Díaz et al. \[2024\]](#) compare the 2010 Maule earthquake to New Zealand’s Canterbury sequence and document that Chilean regional value added remained below its counterfactual path well after the event, while [Benguria \[2022\]](#) reaches a broadly similar conclusion for national GDP. The contemporaneous damage assessment in [ECLAC \[2010\]](#) fixes the scale of the shock—losses near 18% of one year’s GDP, with destruction concentrated in housing and physical infrastructure—and motivates treating the earthquake as a capital shock rather than a demand or institutional shock.

What this strand establishes is the *sign and persistence* of the aggregate response. What it cannot establish, because it works with regional or national aggregates, is the *composition* of that response inside the firm: whether the recovery is driven by rebuilding capital, by rehiring, by changing the skill mix, or by none of these. Opening that black box is the point of using establishment microdata, and it is where this project departs from the aggregate-trajectory literature it otherwise rests on.

2 Disasters and local labor markets

A smaller strand looks past aggregates to the local labor market. [Belasen & Polachek \[2009\]](#) use quarterly establishment data to show that Florida hurricanes raise earnings in directly hit counties—consistent with a local labor-demand pull from reconstruction net of any supply loss—while depressing them in neighboring counties. [Notowidigdo \[2020\]](#) provides the organizing frame for thinking about who bears a *local* demand shock at all: because less-skilled workers are less geographically mobile but better insured by transfers, the welfare incidence of a local shock is distributed differently across skill groups than a naive wage regression would suggest. This is the literature closest in spirit to the project’s wage analysis, and it supplies two cautions that the empirical design has to take seriously. First, a wage response in the hit region can reflect either a demand pull or a compositional change in who remains—so a labor-supply (out-migration) confound must be ruled out before any wage movement is read as demand. Second, the relevant skill groups may be integrated across a wider labor market than the treated region, which blunts the local skill-premium response one might otherwise expect.

The project takes both cautions on board: it uses a household survey (CASEN) to test directly whether the unskilled population fell in harder-hit regions (it did not), and it interprets the fragility of the skill-premium estimate through the lens of geographic integration rather than as evidence against complementarity. What this strand does not do—again for want of capital data—is connect the labor-market response to what is happening on the capital side of the same establishments. That connection is the project’s contribution.

3 Capital–skill complementarity and the skill premium

The motivating hypothesis comes from the capital–skill complementarity literature. [Krusell et al. \[2000\]](#) show that if equipment capital is more complementary with skilled than with unskilled labor, then the secular fall in equipment prices can account for much of the rise in the skill premium. Read as a prediction about shocks, this says that a surge in equipment

capital should raise the relative demand for, and hence the relative wage of, skilled workers. The earthquake offers a natural test: reconstruction is, mechanically, a pulse of new capital.

Lewis [2011] complicates the prediction in a way that turns out to matter for the results. Using U.S. local labor markets, he shows that factor supplies shape technology adoption in equilibrium—an inflow of less-skilled labor slows the adoption of automation machinery—so the observed relationship between capital and the skill premium is an equilibrium object, not a structural parameter that a reduced-form regression recovers. The implication for this project is that a reduced-form “Post $\times$ intensity” coefficient on the skill premium identifies a *net* incidence, not the structural complementarity parameter, and that a small or fragile estimate is consistent with—rather than a refutation of—complementarity once local factor supplies and cross-regional integration are allowed to respond. The project’s finding (a skill-premium response that does not survive the parallel-trends sensitivity analysis) is reported in exactly these restrained terms.

4 Incidence: who captures the returns to capital

If reconstruction raises returns to capital, the natural question is who ends up with them. The incidence literature has answered the analogous question for *fiscal* shocks to the price of capital and labor. Suárez Serrato & Zidar [2016] estimate that the burden of U.S. state corporate taxes is split among firm owners, workers, and landowners, with a substantial share falling on owners; Fuest et al. [2018] find that roughly half of the German local business-tax burden is borne by workers through lower wages; and Gruber [1997], studying Chile’s own payroll-tax reductions, finds the change almost fully shifted into wages with no employment effect. These papers share a design—a tax wedge that moves the cost of a factor—and a question—what share of the change shows up in wages versus profits.

The 2010 earthquake is the same incidence question with a different instrument: the shock is *physical* rather than fiscal. It destroys capital and finances its replacement, and one can ask what share of the reconstruction shows up as higher labor compensation versus as rebuilt capital and recovering returns. The project answers this directly by decomposing the post-shock response across investment, the wage bill, labor’s share of output, and a profit proxy on a common footing. The finding—investment and the total wage bill rise while labor’s share does not—places the incidence on capital, and is, to my knowledge, the first establishment-level incidence decomposition of a physical capital-destruction shock. The fiscal incidence literature supplies both the question and the accounting; the physical shock supplies the variation.

5 Inference: continuous-treatment difference-in-differences

The project’s identifying variation is continuous (ground-shaking intensity by region) and its panel has few clusters (about fifteen regions) and a known pre-trend (central regions were already growing faster). Three methodological papers govern how the results are reported. [Roth \[2022\]](#) shows that pre-testing for parallel trends and conditioning on passing the test distorts inference and has low power against exactly the trends that matter; the project therefore does not treat a passed pre-trend test as a license. [Rambachan & Roth \[2023\]](#) provide the constructive alternative used throughout: instead of assuming parallel trends, bound the post-period bias by the magnitude of pre-period violations (relative-magnitudes and smoothness restrictions) and report the breakdown value at which the result disappears. This is what downgrades the investment effect from “significant” to “robust but suggestive” and what collapses the skill-premium effect. Finally, [Cameron et al. \[2008\]](#) supply the wild cluster bootstrap used to keep inference honest with so few clusters. Together these tools are the reason the paper’s claims are stated conservatively rather than at face-value significance.

6 Where this project sits

Each literature answers part of the question and stops at the edge of its data. The aggregate-trajectory papers establish that the shock is real and persistent but cannot see inside the firm. The local-labor-market papers see wages and employment but not capital. The complementarity papers predict a skill-premium response but, after [Lewis \[2011\]](#), warn that a reduced form will not recover it cleanly. The incidence papers supply the right accounting but for fiscal, not physical, shocks. This project’s contribution is to bring establishment-level capital and labor data to bear on all four at once: it shows that the only margin that moves is reconstruction investment, that output, employment, productivity, the skill premium, and labor’s share do not, and that the incidence of the rebuilding therefore falls on capital rather than labor—reconstruction without reallocation. The methodological literature is what licenses stating that conclusion with the appropriate caution.

References

- Aksoy, C. G., Chupilkin, M., Koczan, Z., & Plekhanov, A. (2024). Unearthing the economic and social consequences of earthquakes. *IZA Discussion Paper No. 17108*.
- Belasen, A. R., & Polachek, S. W. (2009). How disasters affect local labor markets:

- The effects of hurricanes in Florida. *Journal of Human Resources*, 44(1), 251–276. [doi:10.3368/jhr.44.1.251](https://doi.org/10.3368/jhr.44.1.251)
- Benguria Garibaldi, S. (2022). Shaky growth: Chile’s earthquake and its effect on GDP. Master’s thesis, Stockholm School of Economics.
- Cameron, A. C., Gelbach, J. B., & Miller, D. L. (2008). Bootstrap-based improvements for inference with clustered errors. *Review of Economics and Statistics*, 90(3), 414–427. [doi:10.1162/rest.90.3.414](https://doi.org/10.1162/rest.90.3.414)
- Díaz, D., Paniagua, P., & Larroulet, C. (2024). Earthquakes and the wealth of nations: The cases of Chile and New Zealand. *Working paper*, arXiv:2405.12041.
- Economic Commission for Latin America and the Caribbean. (2010). *The Chilean Earthquake of 27 February 2010: An Overview*. Santiago: ECLAC.
- Fuest, C., Peichl, A., & Siegloch, S. (2018). Do higher corporate taxes reduce wages? *American Economic Review*, 108(2), 393–418. [doi:10.1257/aer.20130570](https://doi.org/10.1257/aer.20130570)
- Gruber, J. (1997). The incidence of payroll taxation: Evidence from Chile. *Journal of Labor Economics*, 15(S3), S72–S101. [doi:10.1086/209877](https://doi.org/10.1086/209877)
- Krusell, P., Ohanian, L. E., Ríos-Rull, J.-V., & Violante, G. L. (2000). Capital-skill complementarity and inequality. *Econometrica*, 68(5), 1029–1053. [doi:10.1111/1468-0262.00150](https://doi.org/10.1111/1468-0262.00150)
- Lewis, E. (2011). Immigration, skill mix, and capital-skill complementarity. *Quarterly Journal of Economics*, 126(2), 1029–1069. [doi:10.1093/qje/qjr011](https://doi.org/10.1093/qje/qjr011)
- Notowidigdo, M. J. (2020). The incidence of local labor demand shocks. *Journal of Labor Economics*, 38(3), 687–725. [doi:10.1086/706048](https://doi.org/10.1086/706048)
- Rambachan, A., & Roth, J. (2023). A more credible approach to parallel trends. *Review of Economic Studies*, 90(5), 2555–2591. [doi:10.1093/restud/rdad018](https://doi.org/10.1093/restud/rdad018)
- Roth, J. (2022). Pretest with caution: Event-study estimates after testing for parallel trends. *American Economic Review: Insights*, 4(3), 305–322. [doi:10.1257/aeri.20210236](https://doi.org/10.1257/aeri.20210236)
- Suárez Serrato, J. C., & Zidar, O. (2016). Who benefits from state corporate tax cuts? *American Economic Review*, 106(9), 2582–2624. [doi:10.1257/aer.20141702](https://doi.org/10.1257/aer.20141702)